TECHNICAL SERVICE GUIDE

Managed Disaster Recovery Services

A practical guide to recovery objectives, replication, failover, failback, and managed operations

Translate the supplied SOW into architecture decisions, implementation steps, scope boundaries, and readiness checks.

RPO | RTO | REPLICATION | FAILOVER | FAILBACK | DRILLS

This guide explains the service in technical language with examples for physical, virtual, cloud, endpoint, and Microsoft 365 recovery scenarios.

Prepared from the supplied SOW - Managed DR Services by NxtGen. Exact workloads, dependencies, recovery objectives, capacity, responsibilities, and commercial terms remain subject to the approved statement of work.

01 | READ THE SOW CORRECTLY

Managed DR is a recovery operating model, not only a backup job

The SOW describes a managed disaster-recovery service built around continuous byte-level replication, a NxtGen Tier-3 target site, orchestrated recovery, DR drills, monitoring, and 24x7 operations. The objective is to restore a usable service with predictable evidence, not simply to store copies of data.

The three recovery questions

Question	Technical meaning	Example
How much data can we lose?	Recovery Point Objective, or RPO, describes the maximum acceptable data loss measured in time.	An RPO of seconds means the recovery copy should be close to the latest committed changes, subject to bandwidth and change rate.
How quickly must service return?	Recovery Time Objective, or RTO, describes the target time to restore the required service after activation.	An RTO in minutes requires prebuilt capacity, ordered recovery, tested runbooks, and reachable dependencies.
What must recover together?	Application consistency means recovering databases, application tiers, identity, network paths, and data in the correct dependency order.	Recover the database before the application layer, then validate connectivity, authentication, and business transactions.

The practical interpretation

- Continuous replication reduces the gap between primary data and the DR copy, but the achievable RPO depends on bandwidth, change rate, and workload behavior.
- Failover orchestration can automate a sequence, but activation may still require an authorized human decision or manual trigger.
- A DR drill is evidence that the design, runbook, access, people, and dependencies work together.

Source boundary: the SOW includes ambitious RPO and RTO language, but the exact outcome is workload-specific and requires fine tuning after deployment. Treat seconds and minutes as stated targets, not unconditional guarantees.

02 | CURRENT-STATE CHALLENGES

Design around downtime, data loss, expertise, and complexity

The SOW identifies unplanned outages, weak or untested DR strategies, high RPO and RTO, limited in-house expertise, compliance pressure, and the cost of maintaining a secondary site. These concerns become design requirements.

Challenge	Failure mode	Managed DR response
Unplanned downtime	System failure, ransomware, or human error leaves a critical service unavailable.	Replicate workloads, maintain recovery runbooks, and orchestrate recovery to a prepared DR environment.
Untested recovery	A plan exists but fails when people, access, dependencies, or capacity are tested under pressure.	Run non-disruptive drills, record results, tune runbooks, and track readiness.
High RPO or RTO	Backups are too old or recovery takes too long for the business process.	Tune replication, bandwidth, capacity, application sequencing, and recovery procedures to the required objective.
Limited DR expertise	Internal teams cannot monitor, troubleshoot, or execute the full DR lifecycle consistently.	Use NxtGen-managed deployment, monitoring, incident response, drills, and reporting within the agreed scope.
Cost and complexity	A second site, storage, hardware, software, and skilled operations team are expensive to maintain.	Use NxtGen DR infrastructure and a managed operating model while retaining clear customer responsibilities.

Example: a revenue application

- The business can tolerate only a few seconds of data loss and minutes of service interruption.
- The database, application servers, identity, DNS or routing, and external integrations must be recovered in the correct order.
- The team needs an approved failover decision, an executable runbook, and evidence from a non-disruptive drill.
- The target is not achieved by replication alone; it requires bandwidth sizing, target compute and storage, access, network isolation, and application validation.

03 | PROPOSED ARCHITECTURE

Continuous replication into a prepared NxtGen DR site

The SOW presents a cross-platform design for physical, virtual, and cloud workloads, using continuous byte-level replication to a NxtGen Tier-3 data center with 24x7 NOC management and regular readiness reporting.

Component	SOW description	Technical question to validate
Protected environments	Physical, virtual, and cloud workloads are identified as supported environments.	Which operating systems, hypervisors, cloud platforms, databases, agents, and workload versions are supported?
Replication	Real-time, continuous, byte-level replication is described as the core protection method.	What data, disks, volumes, snapshots, metadata, and application state are replicated, and how is consistency handled?
Target DR site	NxtGen Tier-3 data centers provide the target location in the SOW.	Which site, tenancy, compute, storage, network, isolation, access, and capacity are included?
Management	NxtGen NOC provides 24x7 management, monitoring, alerts, and support.	Which events are monitored, who approves failover, and what is the escalation and response model?
Reporting	DR drills, SLA monitoring, audit-ready reports, dashboards, and monthly health reports are described.	What metrics, evidence, retention, recipients, and reporting cadence are required?

Architecture layers

- Data layer: continuous replication, backup retention, encryption, versioning, and integrity validation.
- Compute layer: target resources at the DR site sized for the agreed recovery workload and tier.
- Network and identity layer: VPN, MPLS, SD-WAN, IAM, firewall rules, routing, and secure DR tunnels.
- Orchestration layer: application priority, dependency order, runbooks, rollback, failover, and failback.

A target site is not automatically a running application environment. Confirm which compute, storage, licensing, network, and application services are pre-provisioned and which are customer-coordinated.

04 | CONFIGURATION

Build recovery readiness in stages

The SOW describes a sequence from assessment through ongoing monitoring. Each stage creates an input for the next; skipping readiness work usually appears later as a failed drill or an unachievable RPO or RTO.

Stage	Technical work	Output to capture
Pre-assessment	Analyze infrastructure, define RPO and RTO, validate connectivity, and size bandwidth.	Workload inventory, change rate, dependency map, link capacity, objectives, and assumptions.
Agent setup	Install agents and configure continuous replication policies.	Agent status, protected disks or volumes, policy, encryption, retention, and alert baseline.
DR site provisioning	Coordinate target compute and storage allocation, define runbooks, and establish network isolation controls.	Target capacity, tenancy, network zones, access, owners, and customer-provided prerequisites.
Failover and failback planning	Configure forward and reverse replication, recovery order, rollback, and cutover windows.	Runbooks, application tiers, dependencies, validation steps, rollback path, and approval matrix.
DR drill and validation	Conduct test failovers, validate results, and document DR readiness.	Observed RPO and RTO, test evidence, issues, remediation owners, and retest date.
Ongoing monitoring	Run 24x7 replication health checks, alerts, and periodic tuning.	Health dashboard, incident records, trend reports, SLA evidence, and tuning decisions.

Customer responsibilities called out in the SOW

- Coordinate compute and storage allocation at the NxtGen data center according to the required scope.
- Define applicable runbooks and approve the recovery sequence and operating decisions.
- Ensure appropriate network isolation controls and provide required connectivity, access, and application dependencies.

Managed service does not mean customer-free service. The best result comes from an explicit responsibility matrix covering NxtGen, customer IT, application owners, network, identity, and third parties.

05 | FAILOVER

Activate recovery with an approved, ordered runbook

Failover moves service operation from the primary environment to the DR environment. The SOW describes disaster events, planned maintenance, and on-demand DR drills as triggers, with preconfigured orchestration and application-aware sequencing.

Failover step	What happens	Evidence or decision
Trigger and declare	A disaster, planned maintenance event, or drill requires recovery activation.	Incident ID, authority to activate, scope, start time, and customer communications.
Assess last good state	Confirm replication health, lag, available recovery point, and target capacity.	Replication status, selected recovery point, data-loss estimate, and readiness check.
Start orchestration	Use the preconfigured runbook to start target resources and recovery actions.	Runbook version, operator or approval, step status, and exception handling.
Recover in dependency order	Bring up identity, database, middleware, application, and access layers in the designed sequence.	Dependency checks, service health, route changes, authentication, and application smoke tests.
Route users and applications	Move access to the DR site through VPN, IP redirect, DNS, or the agreed network method.	Routing change, endpoint reachability, user access, and external integration validation.
Validate and communicate	NxtGen and customer teams validate service readiness and performance at the DR site.	Observed RTO, business transaction test, incident update, and go-forward decision.

Manual decision versus automated sequence

- The SOW describes manual failover orchestration in the deployment plan and automated switchover using preconfigured runbooks in the failover section.
- Interpret this as an authorized activation decision followed by automated or orchestrated execution of the approved steps.
- Confirm which steps are automatic, which require approval, and which require an application owner or customer operator.

Example: a database outage is declared a DR event. The incident commander approves activation, the runbook starts the database and application tiers, routing is changed, and the business owner validates a critical transaction before service is declared operational.

06 | FAILBACK

Return to primary without losing the DR delta

Failback returns service from the DR site to the primary site after the primary environment is stable and synchronized. It is a planned cutover, not simply a reverse button.

Failback step	Technical action	What to validate
Restore primary stability	Confirm the primary site is repaired, reachable, secure, and able to receive the workload.	Infrastructure health, storage, network, identity, dependencies, and capacity.
Reverse replicate	Sync delta changes from the DR site back to the primary site.	Replication direction, lag, conflict handling, last synchronized point, and data integrity.
Plan controlled cutover	Choose a low-impact window and define a rollback or staging option.	Business approval, freeze window, user communication, cutover sequence, and rollback trigger.
Validate application	Bring services back in dependency order and validate data, performance, and integrations.	Application health, business transaction, external connection, and user access.
Complete and report	Run post-failback tests and share logs and validation reports with IT and audit teams.	Final status, achieved recovery objectives, issues, lessons learned, and runbook updates.

Failback example

- The primary storage array is repaired and security controls are revalidated.
- Reverse replication sends all changes made in DR back to the primary site.
- A scheduled cutover stops or quiesces writes, validates the last synchronized point, and starts the primary workload.
- The team confirms application transactions, integrations, monitoring, and user access before closing the incident.

Failback can have a different RTO from failover. Confirm the workload-specific target, the acceptable cutover window, and whether staging or data validation is required before full return.

07 | PROTECTION FEATURES

Combine replication, backup, encryption, versioning, and restore options

The SOW includes both continuous replication and broader data-protection capabilities. Use the right protection mode for the failure scenario instead of treating replication as a replacement for every backup requirement.

Feature	Technical purpose	Example use
Continuous replication	Keep a near-current copy of selected workload data or system state at the DR site.	Recover a critical application after a primary site outage with low RPO.
Secure cloud storage	Encrypt data in transit and at rest using AES-256 as stated in the SOW.	Protect replicated or backed-up data against unauthorized access or storage compromise.
File versioning and archiving	Retain historical versions so corrupted, deleted, or overwritten data can be rolled back.	Restore a clean file version after ransomware or accidental overwrite.
Full-system and file restore	Support full-machine recovery, targeted file restore, or restore to external storage where supported.	Recover a server image after OS failure or restore one deleted file without rebuilding the entire system.
Endpoint and Microsoft 365 backup	Extend managed backup beyond primary application workloads to selected endpoints and Microsoft 365 data.	Recover a user's mailbox or endpoint files after deletion, corruption, or device failure.
Centralized visibility	Use dashboards, health reports, alerts, and monthly reporting to understand protection status.	Find replication lag, failed agents, capacity pressure, or a missed drill before an outage.

Replication is not the same as backup

- Replication is designed for fast recovery and a recent state; corruption or ransomware can replicate into the target.
- Versioning, immutable or protected backup copies, retention, and restore testing support recovery from logical corruption or malicious change.
- Define the recovery point selection method, retention window, access controls, and restore permissions before onboarding.

08 | THREAT COVERAGE

Map failure scenarios to recovery controls

The SOW lists hardware failure, ransomware, human error, natural disasters, corruption, power outages, unauthorized access, application or OS failures, site failure, and compliance readiness. Each scenario requires a different recovery decision.

Scenario	Protection or recovery response	Readiness question
Hardware or power failure	Use replicated data and failover to DR infrastructure while the primary component is repaired.	Can the service start at DR with the required compute, storage, network, and identity dependencies?
Ransomware or malware	Use clean recovery points, versioning, protected backup, encryption, and identity controls.	How is a pre-infection point selected and how do we prove it is clean before release?
Human error or corruption	Use historical versions, integrity checks, targeted restore, or a controlled rollback.	Can we recover one file, one database, or one workload without unnecessarily rolling back everything?
Natural disaster or site failure	Use offsite DR replication and site-level orchestration at the NxtGen target site.	Are people, network routes, access, communications, and external dependencies available from the DR site?
Application or OS failure	Use system image, full-machine restore, or application-aware recovery as supported by the scope.	Does the service scope include application recovery, or only OS boot and infrastructure readiness?
Compliance readiness	Use non-disruptive drills, audit logs, health reports, and evidence of recovery objectives.	Can the organization show the approved plan, test result, exceptions, and remediation record?

Security guardrails

- Use IAM, VPN, firewall rules, and network isolation to restrict who can activate, access, or administer DR resources.
- Protect replication and backup credentials, encryption keys, portal access, and runbook changes.
- Treat a DR copy as production-sensitive data; it may contain the same regulated or confidential information.

The SOW mentions compliance examples including ISO, HIPAA, GDPR, and RBI. Validate which controls, evidence, residency requirements, and customer obligations apply to the specific workload.

09 | DEPLOYMENT AND INTEGRATION

Connect DR to the real operating environment

Deployment is more than installing an agent. It includes topology, bandwidth, authentication, network security, recovery dependencies, runbooks, drills, handover, and ongoing support.

Phase	Technical activity	Deliverable
Assessment and planning	Inventory workloads, dependencies, RPO and RTO, bandwidth, and capacity.	Prioritized workload list, sizing, assumptions, and recovery objectives.
Design and architecture	Design topology, replication paths, network isolation, authentication, target site, and runbooks.	Architecture diagram, dependency map, access model, and approved recovery sequence.
Provisioning	Provision licenses, backup storage, target DR resources, and vaults.	Provisioned DR environment, capacity record, and configuration baseline.
Agent configuration	Install agents and set replication, backup, encryption, schedule, and retention policies.	Protected workload inventory, policy map, health status, and alert baseline.
Network and security integration	Configure VPN, MPLS, or SD-WAN flow, firewalls, IAM, secure tunnels, and access control.	Approved rules, routes, credentials, authentication tests, and isolation evidence.
Testing and handover	Run failover simulations, validate recovery, deliver runbooks, train IT, and define escalation.	Drill report, runbooks, SOPs, dashboards, contacts, and handover acceptance.

Integration questions

- Which identity source and privileged roles control DR portal access and failover approval?
- How are primary and DR networks connected, isolated, and monitored?
- Which DNS, IP redirect, load balancer, VPN, or SD-WAN changes occur during recovery?
- Which customer ITSM or ServiceNow integrations are included, optional, or explicitly out of scope?
- Which third-party licenses, application fixes, and infrastructure purchases remain the customer's responsibility?

10 | MANAGED SCOPE

Make in-scope and out-of-scope boundaries explicit

The SOW includes a useful service boundary. Use it during solution design so that a DR event does not create an unplanned argument about who owns application fixes, network redesign, licensing, or unsupported platforms.

In scope in the SOW	Out of scope in the SOW
Deployment of DR agents and configuration on supported endpoints; continuous replication and data protection.	Diagnosis of issues unrelated to DR configuration, unsupported platforms or legacy systems, and third-party workload licensing.
DR protection for supported physical and virtual workloads; orchestrated failover during events or drills.	Application-specific recovery beyond OS boot, such as database fixes, and root-cause analysis for application-layer issues.
Controlled failback, non-disruptive testing, customized runbooks, and failover workflows.	Long-term backup or archival outside the DR design, customer network redesign, or major re-architecture.
Real-time monitoring and alerting, compliance-ready DR drill documentation, encrypted transmission and storage, and 24x7 support.	Customer ITSM or ServiceNow integration unless explicitly scoped, infrastructure procurement outside NxtGen scope, and unapproved manual interventions.
Access to the DR management and reporting portal.	SLA enforcement for unsupported third-party software and end-user training unless separately agreed.

Responsibility matrix to create

- NxtGen: platform operations, replication monitoring, agreed failover orchestration, reporting, and support.
- Customer IT: workload inventory, access, compute and storage coordination, network isolation, approvals, and application validation.
- Application owners: dependency order, business transaction tests, data validation, and application-specific fixes.
- Third parties: licenses, vendor support, integrations, and components not included in the approved scope.

The out-of-scope list is as important as the feature list. Attach the final scope, assumptions, exclusions, and responsibility matrix to the SOW or implementation plan.

11 | OPERATIONS AND ASSURANCE

Measure recovery readiness before the incident

Managed DR should be operated as a continuous control. Monitoring, drills, health reports, incident records, and improvement actions are what turn a design into dependable recovery.

Operational measure	What it tells you	Example target or evidence
Replication health and lag	Whether the target is receiving changes and how close it is to the intended recovery point.	Dashboard trend, alert history, last synchronized point, and change-rate impact.
Observed RPO	How much data was lost or potentially lost during a drill or event.	Measured gap between primary and recovery point, with workload-specific explanation.
Observed RTO	How long it took to activate, recover, route, and validate the service.	Timeline from activation decision to successful business transaction.
Drill success	Whether people, runbooks, dependencies, access, and capacity worked as designed.	Pass or fail criteria, findings, owners, retest date, and updated runbook.
Protection coverage	Whether all critical workloads, agents, disks, endpoints, and Microsoft 365 sources are protected.	Coverage report, failed-agent list, unsupported assets, and exception approval.
Incident response	Whether alerts become timely decisions and coordinated action.	Escalation records, response time, decision log, communication, and lessons learned.

DR drill checklist

- Confirm drill authorization, scope, observers, customer communications, and stop conditions.
- Validate replication health, target capacity, credentials, network paths, and dependency order.
- Execute recovery without affecting production where the approved test design supports non-disruptive testing.
- Run application and business transaction tests, not just infrastructure boot checks.
- Record observed RPO and RTO, issues, owners, corrective actions, and the next validation date.

12 | IMPLEMENTATION CHECKLIST

Assess, design, confirm, activate, and improve

Use this checklist to convert the SOW into a technically complete managed DR implementation and a repeatable readiness program.

Step	Questions to answer	Evidence to retain
01 Assess	Which workloads, data, users, dependencies, and failure scenarios are critical?	Inventory, data classification, dependency map, current-state risks, RPO and RTO.
02 Design	How will replication, target capacity, network, identity, security, and runbooks work?	Architecture, sizing, topology, policy, access matrix, and responsibility matrix.
03 Confirm	What is included, excluded, customer-provided, measurable, and commercially agreed?	Approved SOW, assumptions, exclusions, SLA definitions, prerequisites, and acceptance criteria.
04 Activate	Can a named authority safely initiate the approved recovery sequence?	Runbook approval, change record, communication plan, and activation decision path.
05 Validate	Does the recovered service meet data, time, security, and business requirements?	Drill report, observed RPO and RTO, application validation, issue log, and retest evidence.
06 Improve	What changed in the workload, threats, dependencies, or business objective?	Health reports, capacity trend, runbook revisions, exception reviews, and next drill plan.

Technical workshop questions

- What is the minimum viable service that must be restored first, and what can wait?
- Which RPO and RTO are required per workload, and what bandwidth and change rate support them?
- Which steps are automatic, which require human approval, and which are out of scope?
- What happens if the primary site is unavailable for failback, or if the recovery copy is corrupted?
- How are DR data, credentials, keys, portals, runbooks, and privileged access protected?
- What evidence must be available for auditors, regulators, customers, and internal risk committees?

Final takeaway: dependable DR is an engineered and exercised system. Replicate the right data, prepare the right capacity, protect the recovery path, orchestrate the right sequence, validate the business service, and improve the runbook after every drill or event.

Source note: this guide is an explanatory interpretation of the supplied SOW - Managed DR Services by NxtGen. Product capabilities, supported platforms, RPO and RTO targets, Tier-3 site details, encryption, SLA terms, scope boundaries, and commercial responsibilities remain subject to the approved statement of work and final confirmation with NxtGen.